INSTRUCTIONS

AF3060 INVERTER DRIVES

INVERTER ASSEMBLY - 331X324EBG01, G04
G02, G05

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62514A

DIAGRAMS

INVERTER CONV. ASM.	ELEM	36C764816EB
	CONN	36D865531EA (G01, G04)
		36D865520EC (G02, G05)
PHASE MODULE	ELEM	36C764814EA
	CONN	36C764815EA
RECTIFIER MODULE ASM.	ELEM	36C765812AA
	CONN	36C764813AA
CURRENT FEEDBACK	ELEM	36C762857AB
PULSE TRANSFORMER	ELEM	36C762856AB
AUXILIARY SUPPRESSION	ELEM	36C762885AE

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

demark of the General Electric Co.

GENERAL ELECTRIC

36C764816EB

CONT ON SHEET 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM

AF-3063 INVERTER ASM

FIRST MADE FOR STANDARD LINE

331X324EAG01,602,604,605

PMA (PHASE MODULE A)
ELEM DIAG 36C764814EA
CONN DIAG 36C764815EA
CAT NO. 331X323EA602

PMB (SAME AS PHASE A)

PMC (SAME AS PHASE A)

AUXILIARY SUPPRESSION ASSEMBLY ASA

RECT (RECTIFIER MODULE)
ELEM DIAG 36C764812AA
CONN DIAG 36C764813AA
CAT NO 331X322AA603

460VAC INPUT

CURRENT FEEDBACK PANEL ASSEMBLY

CURRENT FEEDBACK CARDS
ACF
BCF
CCF

SEE SHEET 2 FOR NOTES AND COMPONENT VALUES

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS	PRINTS TO:	
	5B(3)23	JA(CD)
	5D(BK)	IT(BW)
	5(QC)B	5AE(BW)
	5QC(BK)5	5L(BW)5
	5(TS)50	
	5(S)	

MADE BY C. Mack DEC.20,1974

APPROVALS C. Mack Feb 24, 1975

SPEED VARIATOR DIV OR DEPT

ERIE, PA LOCATION

36C764816EB CONT ON SHEET 2

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

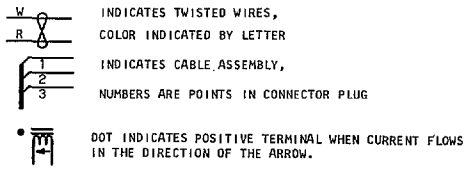
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE NOTES
AF-3063 INVERTER ASM
FIRST MADE FOR STANDARD LINE
331X324EAG01, G02, G04, G05

1. THE ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON KVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.

2. THE L1 CHOKE IS MOUNTED REMOTE TO THE REMAINDER OF THE INVERTER ASSEMBLY. ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.

3. ELEMENTARY WIRING SYMBOLS



W / R — INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER

1 / 2 / 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG

M — DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.

4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.

5. ON PRINTED CIRCUIT CARDS USED IN THIS ASSEMBLY, THE LETTERS "AA" AFTER THE BASIC CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBER AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS "AB", "AC", "AD", ETC. ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARD. AN EARLIER DESIGN CANNOT REPLACE A LATER DESIGN, E.G. AN "AA" CANNOT REPLACE AN "AB".

6. THE NOTED RESISTOR VALUES ON THE CURRENT FEEDBACK CARD ARE DEPENDENT UPON KVA RATING. THE PARTICULAR CARD AND JUMPER CONNECTION IS FOUND IN TABLE I. THE CIRCUIT SHOWN IN DASHED LINES IS A REPRESENTATION OF THE ACTUAL CIRCUITRY AND IS IDENTICAL FOR ACF, BCF AND CCF.

7. RECTIFIER ASSEMBLY AND ASSOCIATED WIRE SHOWN IN DASHED LINES ARE OMITTED FOR G04 & G05.

8. PIR JUMPER SHOWN AS A DASHED LINE IS REQUIRED FOR G01 AND G02.

TABLE I

KVA RATING	L1	C1	C1 & C1	CCA, CCB, CCC	LCA, LCB, LCC	RF RF3	ACF BCF CCF	CURRENT FEEDBACK JUMPER KVA	CONNECTION	ARP ARN
30	4MH	4250MF	2125MF	4MF	90MH	120Ω	193X 199BAG01	10	XA	2500Ω
								15	XB	
								30	XC	
60	3.2MH	8500MF	4250MF	8MF	40MH	56.2Ω	193X 199BAG02	30	XA	1000Ω
								40	XB	
								60	XC	

CONT ON SHEET — DRAWING NO. — SH NO — REV NO. 0

REVISIONS
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.
2 CHG. RF + RF3 ON 15KVA 240 V.C. Gonzago June 25, 1975

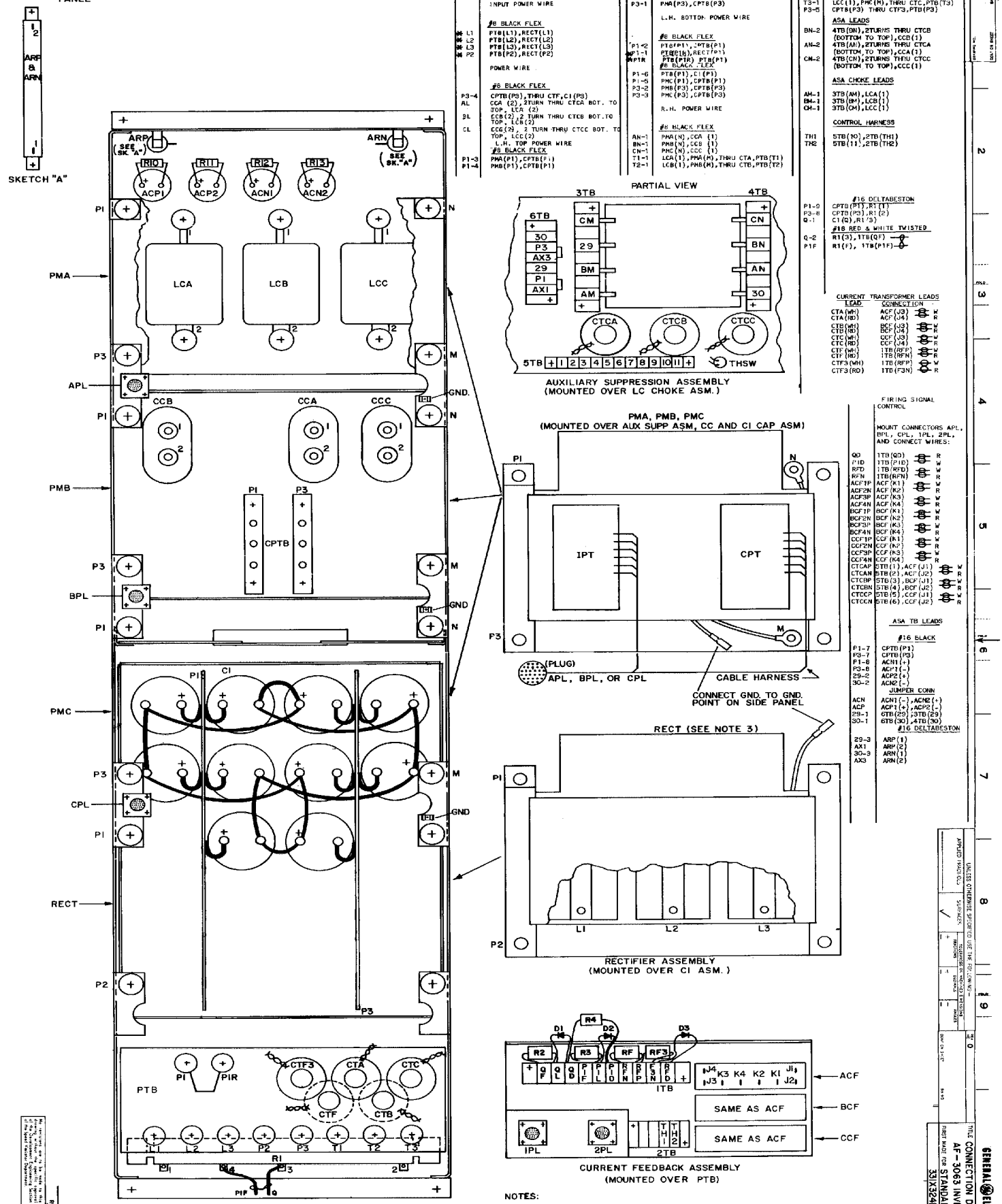
PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	TS(50)
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY G.T. Marsh DEC 26, 1974 — ISSUED G.T. Marsh FEB. 24 1975 — APPROVALS — SPEED VARIATOR DIV OR DEPT ERIE, PA. LOCATION — 36C764816EB CONT ON SHEET FL. SH NO 2

FF-808-F (8-68)



FRONT VIEW OF INVERTER ASM WITH RECTIFIER AND PHASE MODULES REMOVED.

WIRE TABLE

WIRE	CONNECTION
	INPUT POWER WIRE
	#8 BLACK FLEX
L1	PTB(L1),RECT(L1)
L2	PTB(L2),RECT(L2)
L3	PTB(L3),RECT(L3)
P2	PTB(P2),RECT(P2)
	POWER WIRE
	#8 BLACK FLEX
P3-4	CPTB(P3),THRU CTF,CI(P3)
AL	CCA (2), 2TURN THRU CTCA BOT. TO TOP, LCA (2)
BL	CCB(2), 2 TURN THRU CTCB BOT.TO TOP, LCB(2)
CL	CCC(2), 2 TURN THRU CTCC BOT. TO TOP, LCC(2)
	L.H. TOP POWER WIRE
	#8 BLACK FLEX
P1-3	PMA(P1),CPTB(P1)
P1-4	PMB(P1),CPTB(P1)

WIRE TABLE CONT'D

WIRE	CONNECTION
P3-1	PMA(P3),CPTB(P3)
	L.H. BOTTOM POWER WIRE
	#8 BLACK FLEX
P1-2	PTB(P1),CPTB(P1)
P1-1	PTB(P1R),RECT(P1)
P1R	PTB(P1R) PTB(P1)
	#8 BLACK FLEX
P1-6	PTB(P1),CI(P1)
P1-5	PMC(P1),CPTB(P1)
P3-2	PMB(P3),CPTB(P3)
P3-3	PMC(P3),CPTB(P3)
	R.H. POWER WIRE
	#8 BLACK FLEX
AN-1	PMA(N),CCA (1)
BN-1	PMB(N),CCB (1)
CN-1	PMC(N),CCC (1)
T1-1	LCA(1),PMA(M),THRU CTA,PTB(T1)
T2-1	LCB(1),PMB(M),THRU CTB,PTB(T2)

WIRE TABLE CONT'D

WIRE	CONNECTION
T3-1	LCC(1),PMC(M),THRU CTC,PTB(T3)
P3-5	CPTB(P3) THRU CTF3,PTB(P3)
	ASA LEADS
BN-2	4TB(BN),2TURNS THRU CTCB (BOTTOM TO TOP),CCB(1)
AN-2	4TB(AN),2TURNS THRU CTCA (BOTTOM TO TOP),CCA(1)
CN-2	4TB(CN),2TURNS THRU CTCC (BOTTOM TO TOP),CCC(1)
	ASA CHOKE LEADS
AM-1	3TB(AM),LCA(1)
BM-1	3TB(BM),LCB(1)
CM-1	3TB(CM),LCC(1)
	CONTROL HARNESS
TH1	5TB(10),2TB(TH1)
TH2	5TB(11),2TB(TH2)
	#16 DELTABESTON
P1-9	CPTB(P1),R1(1)
P3-8	CPTB(P3),R1(2)
Q-1	CI(Q),R1(3)
	#16 RED & WHITE TWISTED
Q-2	R1(3),1TB(QF)
P1F	R1(F), 1TB(P1F)

PARTIAL VIEW

AUXILIARY SUPPRESSION ASSEMBLY (MOUNTED OVER LC CHOKE ASM.)

CURRENT TRANSFORMER LEADS

LEAD	CONNECTION
CTA(WH)	ACF(J3)
CTA(RD)	ACF(J4)
CTB(WH)	BCF(J3)
CTB(RD)	BCF(J4)
CTC(WH)	CCF(J3)
CTC(RD)	CCF(J4)
CTF(WH)	1TB(RFP)
CTF(RD)	1TB(RFN)
CTF3(WH)	1TB(F3N)
CTF3(RD)	1TB(F3N)

PMA, PMB, PMC
(MOUNTED OVER AUX SUPP ASM, CC AND CI CAP ASM)

FIRING SIGNAL CONTROL

MOUNT CONNECTORS APL, BPL, CPL, 1PL, 2PL, AND CONNECT WIRES:

QD	1TB(QD)
P1D	1TB(P1D)
RFD	1TB(RFD)
RFN	1TB(RFN)
ACF1P	ACF(K1)
ACF2N	ACF(K2)
ACF3P	ACF(K3)
ACF4N	ACF(K4)
BCF1P	BCF(K1)
BCF2N	BCF(K2)
BCF3P	BCF(K3)
BCF4N	BCF(K4)
CCF1P	CCF(K1)
CCF2N	CCF(K2)
CCF3P	CCF(K3)
CCF4N	CCF(K4)
CTCAP	5TB(1),ACF(J1)
CTCAN	5TB(2),ACF(J2)
CTCBP	5TB(3),BCF(J1)
CTCBN	5TB(4),BCF(J2)
CTCCP	5TB(5),CCF(J1)
CTCCN	5TB(6),CCF(J2)

ASA TB LEADS

#16 BLACK

P1-7	CPTB(P1)
P3-7	CPTB(P3)
P1-8	ACN1(+)
P3-8	ACP1(-)
29-2	ACP2(+)
30-2	ACN2(-)

JUMPER CONN

ACN	ACN1(-),ACN2(+)
ACP	ACP1(+),ACP2(-)
29-1	6TB(29),3TB(29)
30-1	6TB(30),4TB(30)

#16 DELTABESTON

29-3	ARP(1)
AX1	ARP(2)
30-3	ARN(1)
AX3	ARN(2)

RECT (SEE NOTE 3)

RECTIFIER ASSEMBLY (MOUNTED OVER CI ASM.)

CURRENT FEEDBACK ASSEMBLY (MOUNTED OVER PTB)

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR G04.
4. FAN ASSEMBLY SUPPLIED WHEN SPECIFIED. SEE SYSTEM DIAGRAMS FOR ACTUAL WIRING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DEC.19, 1974 — SPEED VARIATOR — ERIE, PA.

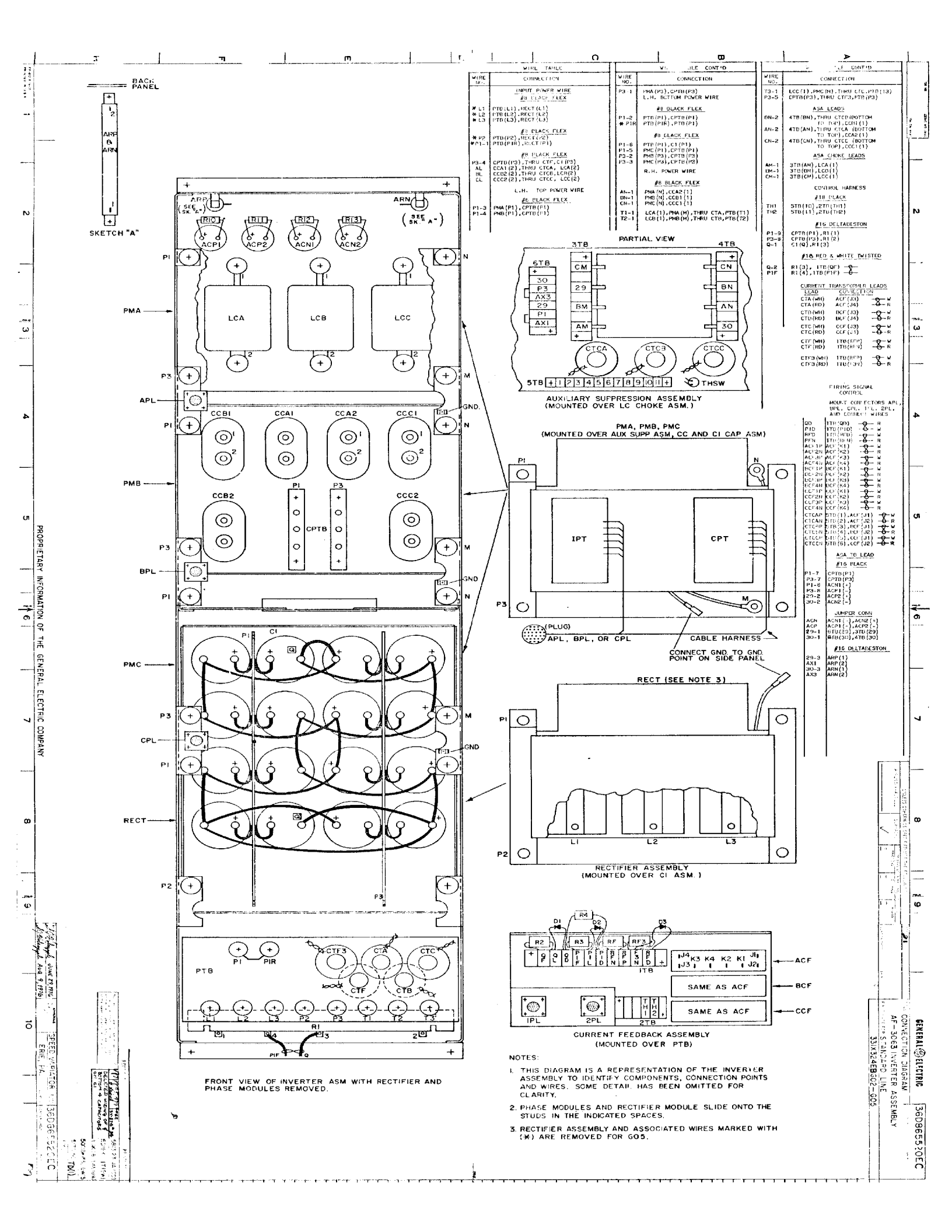
36D86553 E A

REVISIONS

GENERAL ELECTRIC

TITLE CONNECTION DIAGRAM AF-3063 INVERTER ASSEMBLY FIRST MADE FOR STANDARD LINE 33JX324EA901B G04

36D86553EA


WIRE TABLE
WIRE NO.
CONNECTION
INPUT POWER WIRE
#8 FLACK FLEX
* L1 PTB(L1),RECT(L1)
* L2 PTB(L2),RECT(L2)
* L3 PTB(L3),RECT(L3)
#8 BLACK FLEX
* P2 PTB(P2),RECT(P2)
*P1-1 PTB(P1R),RCTF(P1)
#8 BLACK FLEX
P3-4 CPTB(P3),THRU CTF,C1(P3)
AL CCA1(2),THRU CTCA, LCA(2)
BL CCB2(2),THRU CTCB,LCB(2)
CL CCC2(2),THRU CTCC, LCC(2)
L.H. TOP POWER WIRE
#6 BLACK FLEX
P1-3 PMA(P1),CPTB(P1)
P1-4 PMB(P1),CPTB(P1)
WIRE TABLE CONT'D
WIRE NO.
CONNECTION
P3-1 PMA(P3),CPTB(P3)
L.H. BOTTOM POWER WIRE
#8 BLACK FLEX
P1-2 PTB(P1),CPTB(P1)
* P1R PTB(P1R),PTB(P1)
#8 BLACK FLEX
P1-6 PTB(P1),C1(P1)
P1-5 PMC(P1),CPTB(P1)
P3-2 PMB(P3),CPTB(P3)
P3-3 PMC(P3),CPTB(P3)
R.H. POWER WIRE
#8 BLACK FLEX
AN-1 PMA(N),CCA2(1)
BN-1 PMB(N),CCB1(1)
CN-1 PMC(N),CCC1(1)
T1-1 LCA(1),PMA(M),THRU CTA,PTB(T1)
T2-1 LCB(1),PMB(M),THRU CTB,PTB(T2)
WIRE TABLE CONT'D
WIRE NO.
CONNECTION
T3-1 LCC(1),PMC(M),THRU CTC,PTB(T3)
P3-5 CPTB(P3),THRU CTF3,PTB(P3)
ASA LEADS
BN-2 4TB(BN),THRU CTCB(BOTTOM TO TOP),CCB1(1)
AN-2 4TB(AN),THRU CTCA (BOTTOM TO TOP),CCA2(1)
CN-2 4TB(CN),THRU CTCC (BOTTOM TO TOP),CCC1(1)
ASA CHOKE LEADS
AM-1 3TB(AM),LCA(1)
BM-1 3TB(BM),LCB(1)
CM-1 3TB(CM),LCC(1)
CONTROL HARNESS
#18 BLACK
TH1 5TB(10),2TB(TH1)
TH2 5TB(11),2TB(TH2)
#16 DELTABESTON
P1-9 CPTB(P1),R1(1)
P3-9 CPTB(P3),R1(2)
Q-1 C1(Q),R1(3)
#18 RED & WHITE TWISTED
Q-2 R1(3), 1TB(QF)
P1F R1(4),1TB(P1F)
CURRENT TRANSFORMER LEADS
LEAD
CONNECTION
CTA(WH) ACF(J3)
CTA(RD) ACF(J4)
CTB(WH) BCF(J3)
CTB(RD) BCF(J4)
CTC(WH) CCF(J3)
CTC(RD) CCF(J4)
CTF(WH) 1TB(FFP)
CTF(RD) 1TB(FFN)
CTF3(WH) 1TB(FF2)
CTF3(RD) 1TB(F2N)
FIRING SIGNAL CONTROL
MOUNT CONNECTORS APL, BPL, CPL, 1PL, 2PL, AND CONNECT WIRES
QD 1TB(QD)
P1D 1TB(P1D)
RFD 1TB(RFD)
P1N 1TB(P1N)
ACF1P ACF(K1)
ACF2N ACF(K2)
ACF3P ACF(K3)
ACF4N ACF(K4)
BCF1P BCF(K1)
BCF2N BCF(K2)
BCF3P BCF(K3)
BCF4N BCF(K4)
CCF1P CCF(K1)
CCF2N CCF(K2)
CCF3P CCF(K3)
CCF4N CCF(K4)
CTCAP 5TB(1),ACF(J1)
CTCAN 5TB(2),ACF(J2)
CTCBP 5TB(3),BCF(J1)
CTCBN 5TB(4),BCF(J2)
CTCCP 5TB(5),CCF(J1)
CTCCN 5TB(6),CCF(J2)
ASA TO LEAD
#16 PLACK
P1-7 CPTB(P1)
P3-7 CPTB(P3)
P1-8 ACN1(+)
P3-8 ACP1(-)
29-2 ACP2(-)
30-2 ACN2(-)
JUMPER CONN
AGN ACN1(-),ACN2(+)
ACP ACP1(+),ACP2(-)
29-1 BTB(29),3TB(29)
30-1 BTB(30),4TB(30)
#16 DELTABESTON
29-3 ARP(1)
AX1 ARP(2)
30-3 ARN(1)
AX3 ARN(2)
BACK PANEL
ARP & ARN
SKETCH "A"
ARP (SEE SK "A")
ARN (SEE SK "A")
R10
R11
R12
R13
ACP1
ACP2
ACN1
ACN2
PMA
PMB
PMC
CPL
RECT
APL
BPL
GND.
LCA
LCB
LCC
CCB1
CCA1
CCA2
CCC1
CCB2
CCC2
CPTB
PTB
CTF3
CTA
CTB
CTC
CTF
R1
PIF
FRONT VIEW OF INVERTER ASM WITH RECTIFIER AND PHASE MODULES REMOVED.
PARTIAL VIEW
3TB
4TB
6TB
5TB
CTCA
CTC3
CTCC
THSW
AUXILIARY SUPPRESSION ASSEMBLY (MOUNTED OVER LC CHOKE ASM.)
PMA, PMB, PMC (MOUNTED OVER AUX SUPP ASM, CC AND CI CAP ASM)
IPT
CPT
(PLUG) APL, BPL, OR CPL
CABLE HARNESS
CONNECT GND TO GND POINT ON SIDE PANEL
RECT (SEE NOTE 3)
L1
L2
L3
RECTIFIER ASSEMBLY (MOUNTED OVER CI ASM.)
R4
D1
D2
D3
R2
R3
RF
RF3
ITB
ACF
BCF
CCF
SAME AS ACF
SAME AS ACF
IPL
2PL
2TB
CURRENT FEEDBACK ASSEMBLY (MOUNTED OVER PTB)
NOTES:
1. THIS DIAGRAM IS A REPRESENTATION OF THE INVERTER ASSEMBLY TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN OMITTED FOR CLARITY.
2. PHASE MODULES AND RECTIFIER MODULE SLIDE ONTO THE STUDS IN THE INDICATED SPACES.
3. RECTIFIER ASSEMBLY AND ASSOCIATED WIRES MARKED WITH (*) ARE REMOVED FOR 605.
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
GENERAL ELECTRIC
CONNECTION DIAGRAM
AF-3063 INVERTER ASSEMBLY
S-STANDARD LINE
33X324EB302-005
36D965520EC

GENERAL ELECTRIC

36C764814EA

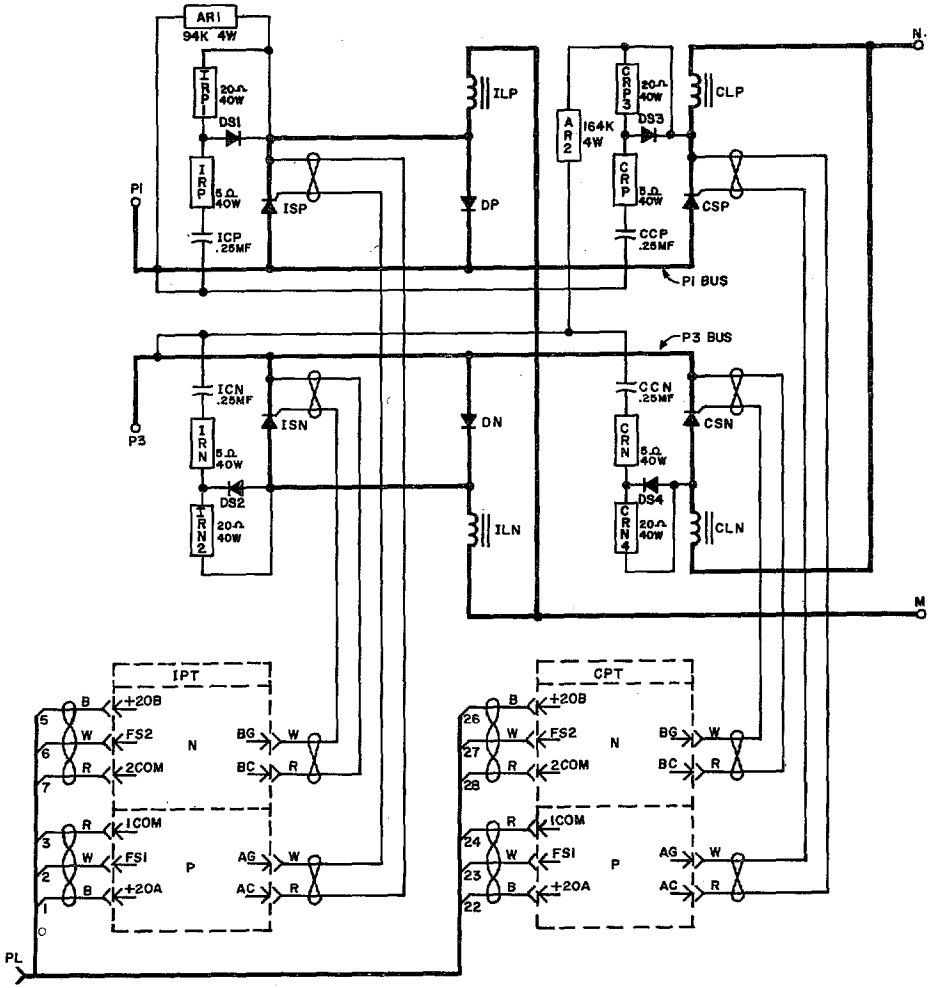
CONT ON SHEET — SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
-	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-3063 PHASE MODULE

FIRST MADE FOR STANDARD LINE
331X323EAG02



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - DS — SNUBBER DIODE
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS.
   - R, W, B — INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER.
   - 1, 2, 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.
   - →)(← — QUICK DISCONNECT "STAB-ON" CONNECTION.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTION OF THIS ASSEMBLY.
5. THE ASSEMBLY CAT. NO. GROUP DISTINCTION IS FOR MAXIMUM RATING:

   G02 — 80 KVA

CONT ON SHEET — SH NO. — DRAWING NO.

REV NO. D 1

REVISIONS

REV			
D	1	Apr 15 76	

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO

5B(S)123	JA(CD)
5D(BK)	
5QC1B	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY R Hayek DEC. 20, 1974
ISSUED F Marek FEB 21 1975

APPROVALS

SPEED VARIATOR — DIV. OR DEPT.
ERIE, PA. — LOCATION

36C764814EA — CONT ON SHEET — SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764815EA

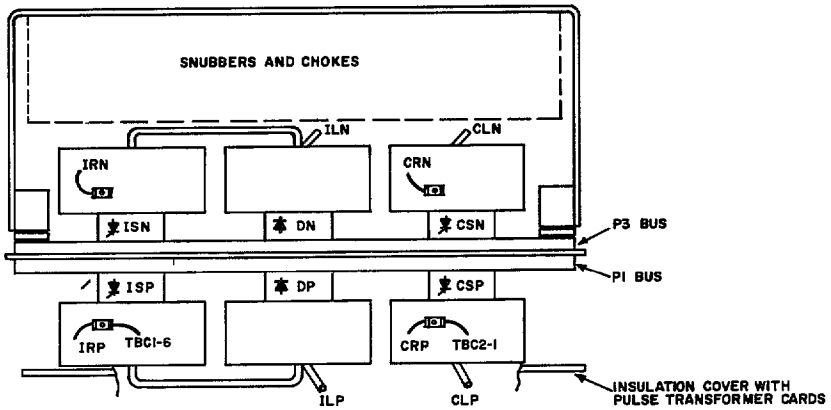
CONT ON SHEET | SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE CONNECTION DIAGRAM
AF-3063 PHASE MODULE

FIRST MADE FOR STANDARD LINE
331X323EA602

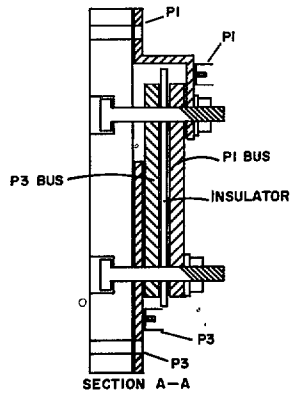


TOP VIEW OF PHASE MODULE

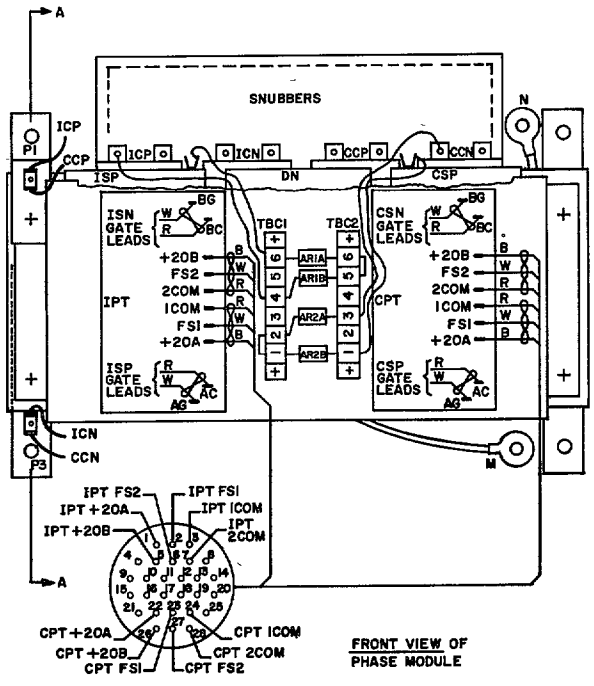
NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE PHASE MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.
2. GATE LEAD AND THE PULSE TRANSFORMER INPUT HARNESS ARE TWISTED WIRES INDICATED BY THE SYMBOL — W, R, B

WIRE COLOR IS INDICATED FOR EASY IDENTIFICATION. GATE LEADS ARE WHITE, CATHODE LEADS ARE RED.

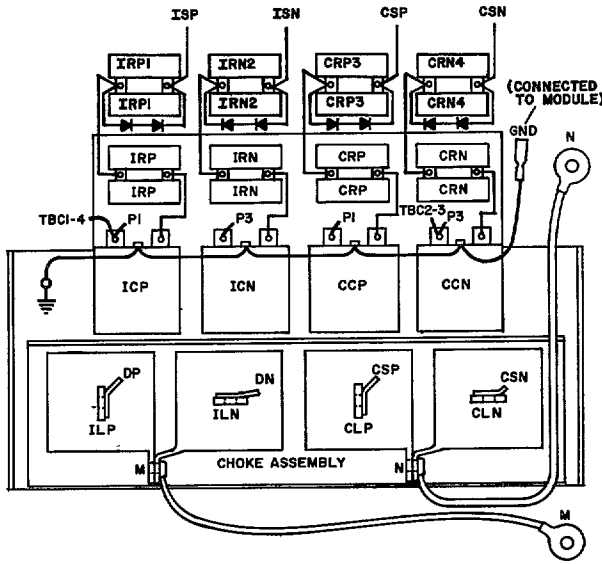


SECTION A—A
VIEW OF BUS ASSEMBLY



PL
(VIEWED FROM WIRE SIDE)

FRONT VIEW OF PHASE MODULE



FRONT VIEW OF SNUBBERS, CHOKES WITH HEATSINK ASSEMBLY REMOVED

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QCBK)5	
5(TS)50	
5(S)	

MADE BY DEC 20, 1974
ISSUED FEB 24, 1975

APPROVALS

SPEED VARIATOR — DIV OR DEPT
ERIE, PA — LOCATION

36C764815EA — CONT ON SHEET — SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-80B-F (4-72) PRINTED IN U.S.A.

GENERAL ELECTRIC

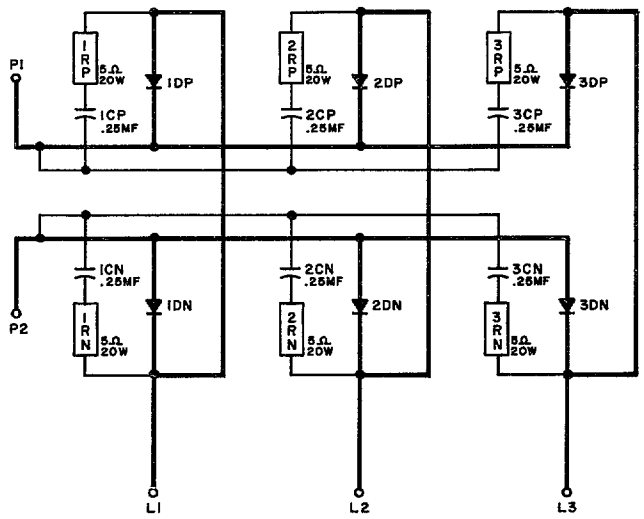
36C764812AA
CONT ON SHEET SH NO

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-3061 RECTIFIER MODULE

FIRST MADE FOR STANDARD LINE
331X322AAG02, G03



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES INDICATING PHASE (1, 2, OR 3) AND SUFFIXES OF P OR N INDICATING POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   C – CAPACITOR
   D – DIODE
   R – RESISTOR
2. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
3. THE ASSEMBLY CAT NO. GROUP DISTINCTION IS FOR MAXIMUM RATINGS:

G02	50 KVA	230AC
G03	60 KVA	460AC

36C764812AA CONT ON SHEET SH NO
REV NO 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC) B	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY J. Mattio SEPT 27, 1972
ISSUED Nov 24, 1971

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION

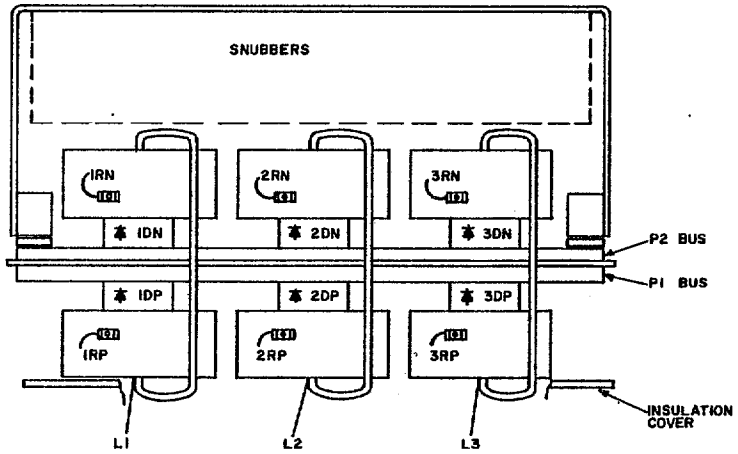
36C764812AA CONT ON SHEET SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

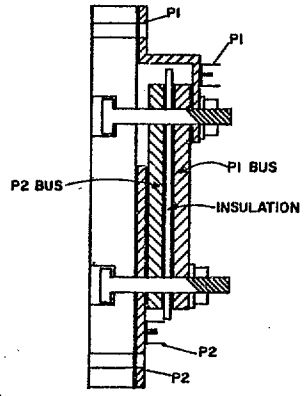
TITLE CONNECTION DIAGRAM
AF-306I RECTIFIER MODULE
FIRST MADE FOR STANDARD LINE
331X322AAG02,G03



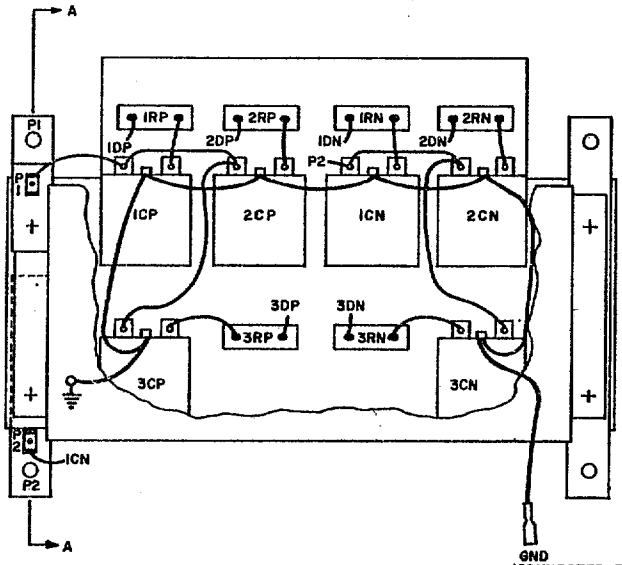
TOP VIEW OF RECTIFIER MODULE

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE RECTIFIER MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.



SECTION A–A
VIEW OF BUS ASSEMBLY



FRONT VIEW OF RECTIFIER MODULE
WITH CUTAWAY SHOWING SNUBBERS

GND
(CONNECTED EXTERNAL TO MODULE)

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS
IRC 91 prod Feb 24, 1975 EHS GPS

PRINTS TO

5B(3)23	JA(CD)
5D(BK)	
5(QC)8	
5QCBK)5	
5(TS)50	
5(S)	

MADE BY G. Mattia OCT. 10, 1972
ISSUED J.D. Keane Nov 24, 1972
ORIGINALS EPS KDP
SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION
36C 764 813 AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

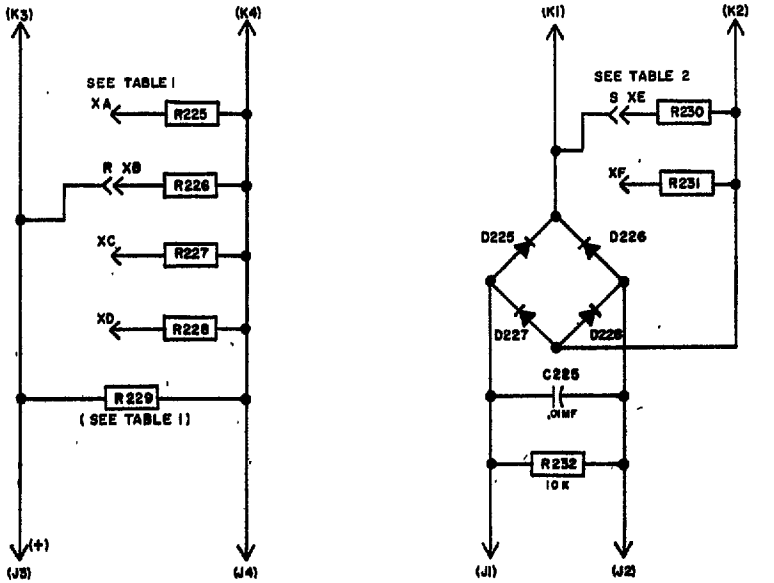
GENERAL ELECTRIC

SCHEMATIC DIAGRAM

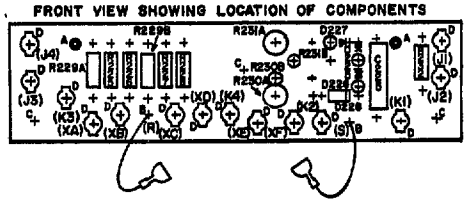
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
.718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
CURRENT FEEDBACK

FIRST MADE FOR STANDARD LINE
193X187ABG01, G02, G03, G04



FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	2
C	.187	3
D	.101	14

NOTE: 1. CARD SIZE, 4.900 $^{+.000}_{-.018}$ X 1.300 $^{+.000}_{-.016}$

NOTES

TABLE I

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE				
	R225(XA)	R226(XB)	R227(XC)	R228(XD)	R229
G01	470Ω	56Ω	33.2Ω	22.1Ω	23.8Ω
G02	1K	39Ω	274Ω	18.2Ω	11.7Ω
G03	68Ω	33Ω	15Ω	10Ω	6.91Ω
G04	10K	39Ω	15Ω	10Ω	5.46Ω

TABLE 2

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE	
	R230(XE)	R231(XF)
G01	63.6Ω	70.6Ω
G02	38.0Ω	49.1Ω
G03	27.0Ω	35.0Ω
G04	21.5Ω	28.2Ω

TABLE 3

GROUP NUMBER	RESISTANCE BETWEEN K3 & K4 USING JUMPER			
	XA	XB	XC	XD
G01	22.70Ω	16.7Ω	13.9Ω	11.5Ω
G02	11.5Ω	8.98Ω	8.19Ω	7.00Ω
G03	7.89Ω	7.05Ω	5.60Ω	4.71Ω
G04	5.47Ω	4.81Ω	4.01Ω	3.54Ω

36C762857AB
REV NO. C

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	9428519AB 004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

5C(B)(K)	5B(C)
5B(2)22	5D(BK)
5(S)	5B(C)
	5B(C)
	BK5(C)
	5B(C)(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY D. Gunnell APR 18 1972
ISSUED D. Gunnell Rev. No. 0 1972

SPEED VARIATOR — DIV OR DEPT
ERIE — LOCATION

SCHEMATIC DIAGRAM

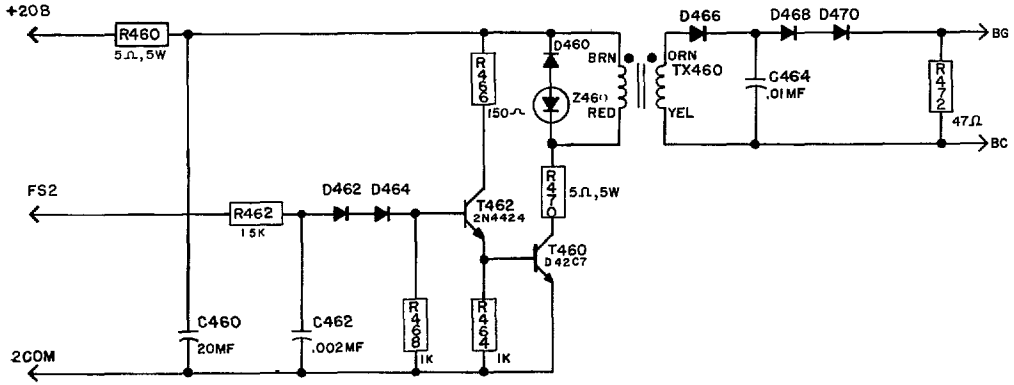
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING

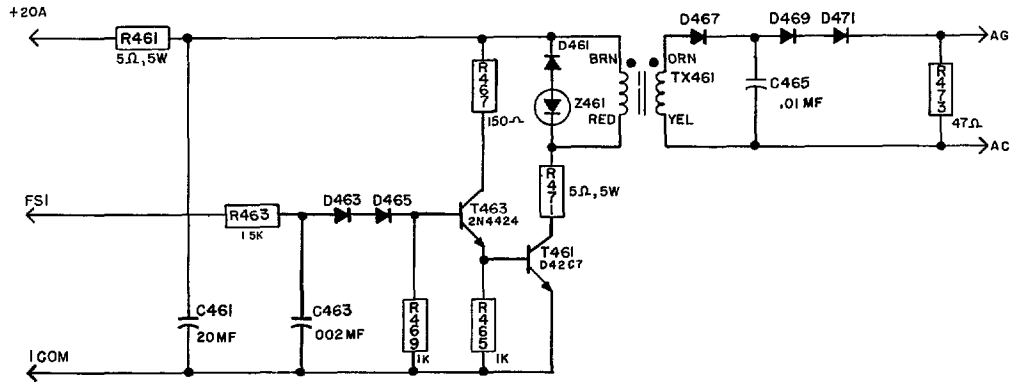
APPLIED PRACTICES	SURFACES	FRACTION	DECIMAL	ANGLES
718A515	✓	±	±	±

GENERAL ELECTRIC

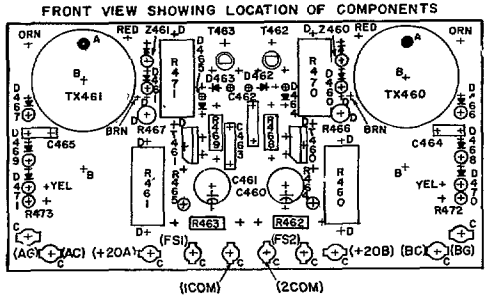
TITLE: PRINTED CIRCUIT DIAGRAM
PULSE TRANSFORMER
FOR STANDARD LINE
193X186ABG01

36C762856AB





FRONT VIEW SHOWING LOCATION OF COMPONENTS



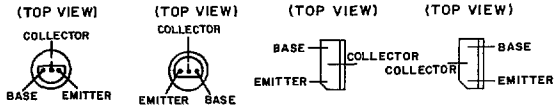
HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QUAN
A	.157	2
B	.187	4
C	.101	10
D	.052	12

NOTE 1 CARD SIZE $5.130 \pm \frac{.000}{.015} \times 2.850 \pm .020$

TRANSISTOR LEAD SKETCHES





SCALE	ETCHED CIRCUIT BOARD DWG	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	942B521AA004	—	—

REVISIONS

No revisions are to be made to this drawing, without the specific approval of the Development Engineering Section of the Speed Variator Department

5QC(BK)5	5 (S)
	5D(BK)
	5B(2)
	5(TS)150
	5(QC)B

R. Novak Jan-25,1972
D.S. Jones JAN 26,1972

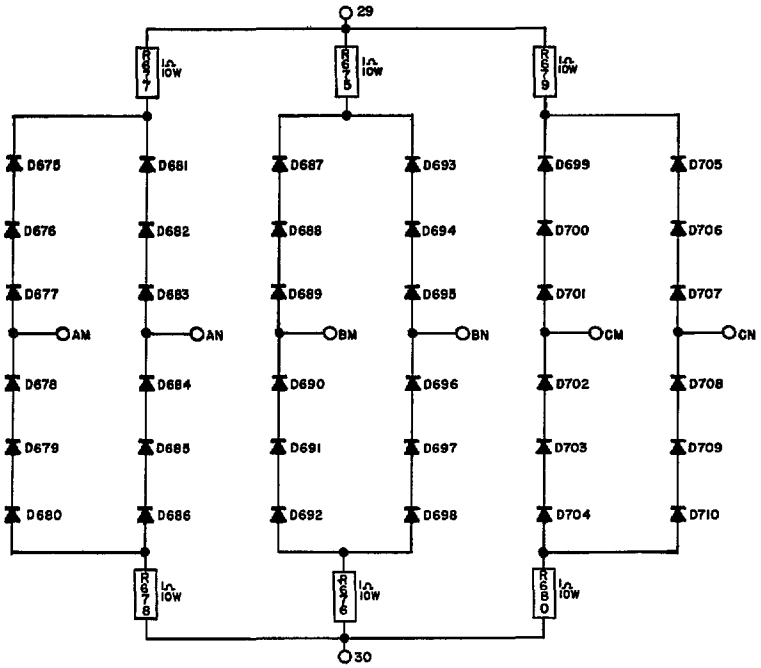
SPEED VARIATOR
ERIE, PA

36C762856AB

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC — 36C7[redacted]35AE

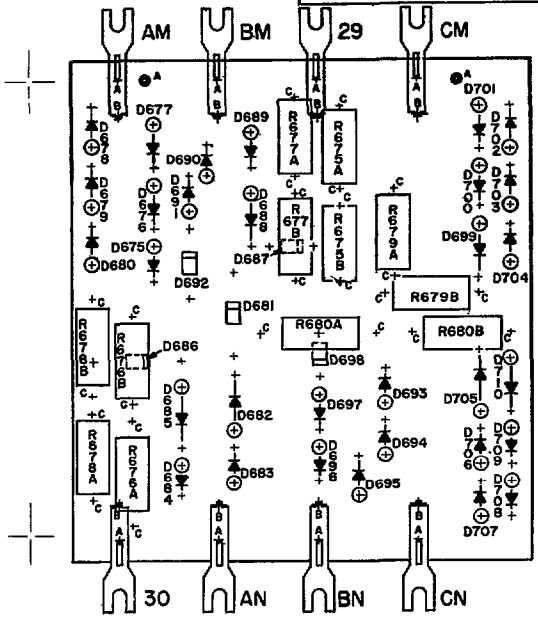
SCHEMATIC DIAGRAM



UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
AUXILIARY SUPPRESSION
FIRST MADE FOR STANDARD LINE
193X193AEG01



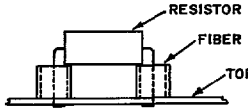
FRONT VIEW SHOWING LOCATION OF COMPONENTS

NOTE
1. CARD SIZE, $5.700^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

HOLE TABULATION

ALL HOLES .062 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC	DIA.	QUAN.
A	.157	10
B	.101	8
C	.052	24



MOUNTING SKETCH FOR RESISTORS
R675A, R675B, R676A, R676B, R677A, R677B, R678A, R678B, R679A, R679B, R680A, R680B

CONT ON SHEET
SH NO
REV NO
0

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	942B604AD004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO

SQC(BW)	5 (S)
IT(BW)	5D(BK)
JA(CD)	582)22
DS(REP)	5L(BW)5
	5(TS)50
	5(QC)8

DRAWN MA Popko MAY 26, 1976
TRACED E. Farese MAY 26, 1976
APPROVALS MB
SPEED VARIATOR ERIE, PA
DIV OR DEPT LOCATION
36C762885AE
CONT ON SHEET SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY